MIL-D-81688A (AS)
1 May 1977
SUPERSEDING
MIL-D-81688 (AS)
29 October 1971

MILITARY SPECIFICATION

DISPLAY SET, MISSION RECORDER AN/UPQ-5B

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification is a Mission Recorder Display Set that will accept input signals from a Video Tape Recorder/Reproducer, Type AN/USH-17(V). The equipment shall display airborne ground map radar presentations, navigation data block quantities, and reproduce aircraft intercommunication system (ICS) audio. Provisions shall be made to obtain instant hard copies of the display information. The equipment shall also include a 400 Hz power source for the Video Tape Recorder/Reproducer.

*1.2 Classification - The equipment covered by this specification shall consist of a single movable console containing the following Weapons Replaceable Assemblies (WRA)s:

Designation	Common Name	Nomenclature	Paragraph
A1	Hard Copy Radar Display (HCRD)	Indicator, Radar IP-1100A/UPQ-5	3.5.1
A2	Data Display	Indicator-Decoder, Digital Data ID-1823A/UPQ-5	3.5.2
A3	Oscilloscope	Oscilloscope AN/USM-296	3.5.3
A4	Radar Display	Indicator, Radar IP-1101A/UPQ-5	3.5.4
A5	Radar Recorder Mount	Mount, Mission Recorder, MT-4557A/UPQ-5	3.5.5
A6	Power Converter	Converter, Frequency, Static CV-2933/UPQ-5	3.5.6
A7	Power Supply A	Power Supply PP-6743/UPQ-5	3.5.7
A8	Power Supply B	Power Supply PP-6744/UPQ-5	3.5.8
A9	Blower	Cooler, Air, Electronic Equipment HD-949/UPQ-5	3.5.9
A10	Equipment Cabinet	Cabinet, Electrical Equipment CY-7319A/UPQ-5	3.5.10

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1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

SPECIFICATIONS

Military

MIL-S-901C 15 January 1963 and Amendment 1 5 September 1963	Shock Tests, H. I. (High-Impact): Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-915C 15 March 1968	Cable, Electrical, Special Purpose, General Specification for
MIL-M-7793C 14 August 1961 and Amendment 2 30 July 1967	Meter, Time Totalizing
MIL-E-16400F 24 February 1966 and Amendment 4 15 May 1968	Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555F 3 July 1968 and Amendment 1 10 January 1969	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303B 1 September 1966	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307C 1 October 1958 and Amendment 4 19 September 1966	Nomenclature and Identification for Electronic; Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-21097B 4 September 1963 and Amendment 3 15 November 1966	Connector, Electrical, Printed Wiring Board, General Purpose, General Specification for

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Naval Air Systems Command

AR-10A 1 January 1969	Maintainability of Avionic Equipment and Systems General Requirements for
AR-34 26 March 1969	Failure Classification for Reliability Testing, General Requirements for

STANDARDS

Military

MIL-STD-167 20 December 1954	Mechanical Vibrations of Shipboard Equipment
MIL-STD-461A 1 August 1968 and Notice 1 7 February 1969	Electronic Interference Characteristics Requirements for Equipment
MIL-STD-749A 1 June 1965 and Notice 1 28 September 1965	Preparation and Submission of Data for Approval of Nonstandard Electronic Parts
MIL-STD-761B 15 July 1965	Electrical Power, Alternating Current for Shipboard Use Characteristics and Utilization of
MIL-STD-781B 15 November 1967 and Change 1 28 July 1969	Reliability Test Exponential Distribution
MIL-STD-785A 28 March 1969	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794 11 March 1969	Parts and Equipment, Procedures for Packaging and Packing of
MS3120D 15 February 1967	Connector, Receptacle, Electric, Crimp-type, Wall Mounting, Flange, No. 4 Holes, Bayonet Coupling
MS17325C 31 December 1969	Meter, Time Totalizing Digital, 115 Volt, 60 or 400 Hertz
MS24266G 14 February 1967	Connector, Plug, Electrical - Straight Miniature

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MS28053F 31 December 1969	Meter, Time Totalizing Digital, 115 Volts, 400 Hertz
MS3102C 15 May 1969	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provisions for preproduction testing.

3.2 Materials, Parts and Processes - In the selection of materials, parts and processes, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered. When used, microelectronic devices shall have two or more sources and the device from each source shall be interchangeable. Interchangeability shall be demonstrated by random installation of devices from two or more sources during fabrication of production quantities. Successful operation of the equipment with a random mixing of device sources shall be proof of interchangeability. Unsuccessful operation of the equipment shall be evaluated to determine if a device is faulty or not interchangeable.

(2) Other parts and materials requirements shall conform to Specification MIL-E-16400.

(3) Nonrepairable subassemblies shall be used in accordance with paragraph 3.1.3(2) of Specification AR-10.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.2.1 Nonstandard Parts and Material Approval - Approval for use of nonstandard parts and materials (including electron tubes, transistors, diodes and microelectronics devices) shall be obtained as outlined in specification MIL-E-16400. Nonstandard parts data shall be in accordance with Step I and Step II only of MIL-STD-749.

3.2.2 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Section 3.5 of this specification.

3.3 Design and Construction - The equipment shall be basically identical in design, construction, workmanship, parts, processes and materials to the prototype units of the AN/UPQ-5 which were designed in accordance with MIL-E-16400 except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 975 pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including built-in-test provisions, shall have 200 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

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Unit	Type of Meter
A1 Hard Copy Radar Display	MS17325-2
A2 Data Display	MS17325-2
A4 Radar Display	MS17325-2
A6 Frequency Converter	MS28053-5
A7 Power Supply A	MS17325-2
A8 Power Supply B	MS17325-2

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-16400.

3.3.3.2 Interconnecting Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-C-915. The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall be supplied as part of the equipment.

*3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of MIL-E-16400. The configuration of all control panels must be approved by the procuring activity prior to Preproduction Testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-16400.

*3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the class IC equipment requirements of MIL-STD-461, Test Methods CE03, CS01, CS02, CS06, RE02, RS01, RS02-2, and RS03. Conducted and radiated emissions from 30 Hz to 5 kHz shall not exceed the limits of CE01 and RE01 by more than 20 dB. Conducted emissions from 5 kHz to 20 kHz shall not exceed the limits of CE01 by more than 15 dB. The equipment shall meet the requirements of test method RE01 for frequencies above 5 kHz. Magnetic interference measurements shall be made at a distance of 24 inches, minimum, from the face of the CRT.

3.3.7 Provisions for Maintainability - Construction and packaging shall be as outlined in paragraphs 3.1.2, 3.1.4 and 3.1.5 of Specification AR-10.

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3.3.7.1 Weight of WRA's - The weights of the individual WRA's shall not exceed the weights specified in section 3.5.

3.3.7.2 Test Points - Test points shall be provided to aid in the maintenance of the equipment. Test points shall be in accordance with section 3.5.

3.3.7.3 Organizational Level Maintenance - Equipment maintenance at the organizational level shall be performed using the built-in-test (BIT) features. After a fault has been detected by the BIT feature, a built-in oscilloscope shall be used to assist in fault isolation to a replaceable assembly (card) level, 90 percent of faults shall be located to the replaceable assembly level by this method. The Mean-Time-To-Fault-Isolation (MTTFI) to a replaceable assembly shall be 30 minutes.

*3.3.7.4 Accessibility - The Hard Copy Radar Display and Data Display WRA's shall be mounted on rotating slides, permitting the WRA's to swing downward. All other units, except the Blower shall be mounted on regular chassis slides.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. Name tags shall be applied down to the WRA level.

3.3.9 Safety - Provisions for personnel and equipment safety shall be in accordance with MIL-E-16400.

3.3.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 95% relative humidity
Input power voltage	115 V, AC ± 5%; 60 Hz, ± 5%

3.3.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental conditions or reasonable combination of these conditions as specified in Specification MIL-E-16400F, and amendment 4, for Class 2 equipment, except as modified herein.

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3.3.11.1 Vibration - The equipment shall be capable of meeting the type I vibration requirements of MIL-STD-167 with exception that vibration in the side to side axis shall not exceed 20 Hz with the top of the cabinet supported.

3.3.11.2 Shock - The equipment shall be capable of withstanding Grade B shock testing requirements of MIL-S-901.

3.3.12 Warm-Up Time - The time requirement for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.13 Input Electrical Power - The equipment shall be supplied with the power cable, terminated with a type UP131M plug.

3.3.13.1 Operating Power - The equipment shall meet all the applicable requirements of MIL-STD-761 and shall give specified performance when energized from the following power source having characteristics and limits as defined in MIL-STD-761. The power required shall not exceed the specified amount.

(1) AC Power (Single Phase), 115V, 60 Hz, Type I, 1500 VA.

3.3.13.2 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failure or poor adjustment shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses and other protective devices shall be held to a minimum.

3.3.13.3 Undervoltage Protection - The equipment shall not be damaged by voltage below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13.4 High Voltage Delay - The equipment shall delay application of high voltage to the anodes of the radar display indicators for 30 seconds after the application of power voltage.

3.3.14 Cooling - Cooling for the equipment shall comply with the requirements of Specification MIL-E-16400 and shall not require the use of specific ducted external cooling. All necessary fans and/or heaters shall be supplied as part of the equipment.

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3.3.14.1 Over Temperature Indicators - Over temperature front panel indicator lamps shall be provided to monitor for excessive temperature of the most critical components of WRA's A1, A4 and A6.

3.3.15 Movability - The equipment shall be designed to permit easy transport.

3.3.15.1 Lifting Eyes - Lifting eyes shall be provided in accordance with MIL-E-16400.

3.3.15.2 Casters - The equipment shall be supplied with four, base mounted, casters for ease of mobility. The casters shall be removable for permanent installations.

3.3.16 Light Shield - An adjustable, removable hood assembly, FSN IR1560 938 9807FA shall be provided by the procuring activity to prevent the impingement of ambient illumination on the face of the display tube. Mounting provisions for the hood shall be provided by the contractor and must be approved by the procuring activity. The time required for installation or removal of the hood assembly (without tools) shall not exceed 10 seconds.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall operate in conjunction with an Airborne Video Tape Recorder/Reproducer, type AN/USH-17(V) and shall provide the necessary interface to display a ground map radar presentation, data block quantities and reproduce ICS audio. Means shall also be provided for obtaining an instant "hard copy" of the radar display and selected data block quantities.

3.4.2 Input signals - The equipment shall be designed to accept the following input signals from the AN/USH-17(V):

*3.4.2.1 Horizontal Sweep Input - The equipment shall be capable of displaying a 60 μ second ramp with a ramp delay not to exceed 1.0 μ second. The input signal shall not exceed plus or minus 1.3 volts. Zero input shall correspond to the center of the horizontal axis. An input voltage of ±0.7 volt shall correspond to a nominal ±3.0 inch vertical deflection on the radar display. The horizontal sweep source impedance shall be 70 ohms, maximum. The allowable sweep recovery time shall be greater than 20 μ seconds.

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*3.4.2.2 Vertical Sweep Input - The equipment shall be capable of displaying an 18-μ second ramp with a ramp delay not to exceed 1.0 μ second. The input signal shall not exceed plus or minus 1.3 volts. Zero input shall correspond to the center of the vertical axis. An input voltage of ±1.1 volts shall correspond to a nominal ±3.0 inch vertical deflection on the radar display. The vertical sweep source impedance shall be 70 ohms, maximum. The allowable sweep recovery time shall be greater than 20 μ seconds.

3.4.2.3 Intensity Gate Input - The intensity gate input shall be compatible with TTL logic levels. A logic "1" corresponds to an unblanking signal while a logic "0" requires that the radar display be blanked.

3.4.2.4 Composite Video - The composite video shall be an AC-coupled signal that can reach a positive one-volt level and has a negative one-half volt 5 μ second pulse at the end of the video pulse. Video bandwidth shall be 5 MHz. The video signal contains the azimuth cursor as well as the range marks and range cursor. Video source impedance shall be 75 ohms.

3.4.2.5 ICS Audio Input - The equipment shall be capable of reproducing ICS audio. The nominal audio level shall be 1 volt peak-to-peak and the source impedance shall be 600 ohms.

3.4.2.6 Data Block Input - The data block signal input shall consist of a 30 bit data segment. The data input shall have the characteristics as shown in Figure 1.

3.4.3 Radar Display - The equipment shall be capable of displaying either PPI or B-Scan radar presentations. The display shall have the following characteristics:

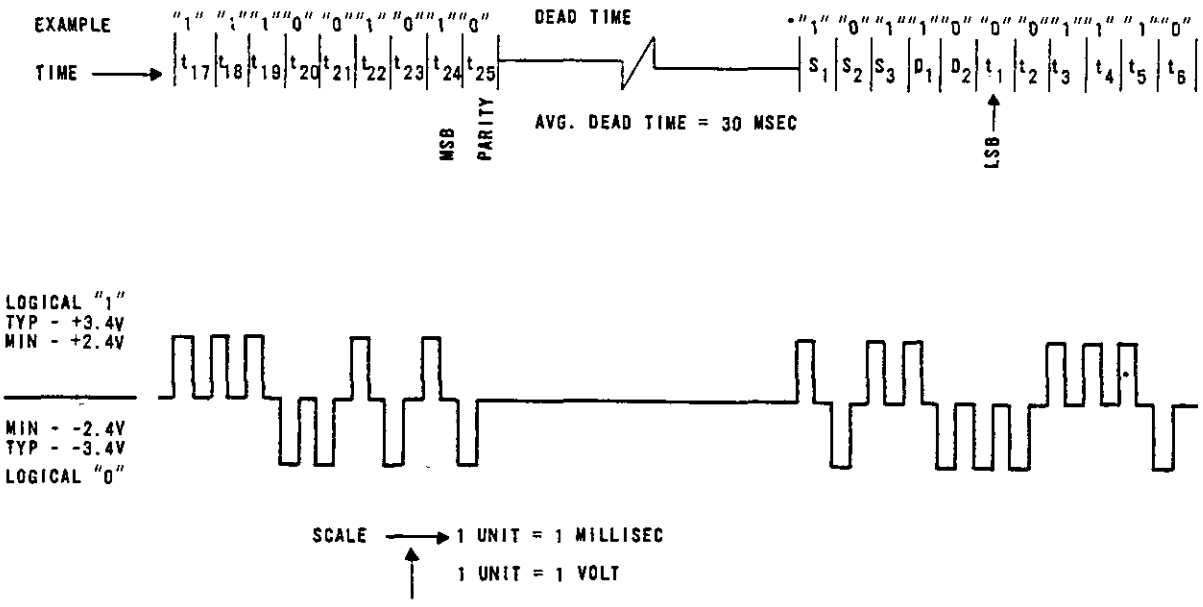
3.4.3.1 Indicator Tube - The indicator tube shall be a seven inch cathode ray tube, type 7M111P14M, electromagnetically deflected and electrostatically focused. The phosphor shall be P14 type. A removable amber filter shall be provided.

3.4.3.2 Brightness - The maximum display tube brightness shall not be less than 5 foot-lamberts when measured on the face of the indicator tube.

3.4.3.3 Spot Size - The indicator tube spot size at a brightness level of 5 foot-lamberts or less shall not exceed 0.007 inches at any point on the face of the tube.

3.4.3.4 Magnetic Shield - A high-permeability magnetic shield shall be provided to protect the display tube from external fields.

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SYSTEM CLOCK $\approx$ 500 Hz

$S_1 \rightarrow S_3$ PATTERN RECOGNIZE

D_1, D_2 RADAR MODES. ARE 30. AND ARE 60 RESPECTIVELY

$t_1 \rightarrow t_8$ ADDRESS

$t_9 \rightarrow t_{24}$ DATA

t_{25} PARITY

SYNC TIME (3 BITS) $\approx$ 6 MSEC

WORD TIME (27 BITS) $\approx$ 54 MSEC

DEAD TIME (VARIABLE) $\approx$ 30 MSEC NOMINAL

NOTE: DISPLAY TERM ADDRESS AND SCALING ARE REQUIRED TO BE VARIABLE.

FIGURE 1. DIGITAL DISPLAY.

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3.4.3.5 Linearity - Display non-linearity introduced by the indicator shall not exceed 1.5 percent.

3.4.3.6 Tonal Renditions - The dynamic range of display brightness shall not be less than 15 dB. This is equivalent to a tonal rendition of five grey shades. Suitable gamma correction shall be provided, if necessary, to meet this requirement.

3.4.3.7 Protective Circuitry - Circuitry shall be provided to protect the cathode ray tube from damage due to a loss of sweep voltage.

3.4.4 Data Block and Data Freeze Processor - The equipment shall perform all the necessary signal conditioning, decoding scale factor adjustment and binary to BCD conversion of the digital data block and data freeze quantities.

*3.4.4.1 Data Block Display - Each data block quantity shall be individually displayed in numeric form. Provisions for 15 primary data block displays shall be provided. The displays shall be continuously updated and shall contain the data block quantities identified in Table I. Data block quantities 1 through 6 of Table I shall be displayed adjacent to the radar display's CRT as shown in Figure 2.

3.4.4.2 Data Freeze Display - The data freeze display shall consist of a single numerical display. The quantity to be presented shall be selected by a panel control switch. Receipt of a weapon release signal indicates that data freeze quantities have been recorded and are available for display and shall light a weapons release indicator lamp. A manual reset shall be provided to clear the data freeze storage register and turn off the weapons release indicator lamp. The Data Freeze quantities to be displayed and the relative panel control switch position shall be in accordance with Table II. An identification plate, adjacent to the panel control switch, shall identify the quantity to be presented on this display at each switch position.

3.4.4.3 Parity - A parity error light shall be provided.

3.4.5 Audio - The equipment shall have an amplifier speaker system for reproduction of ICS audio. The audio shall be supplied to a front panel headphone jack type, JJ-089. The audio level shall be controlled by a front panel volume control.

*3.4.6 Hard Copy - Hard copies of the radar display shall be provided by an externally mounted, modified, Hewlett Packard Oscilloscope Camera, model 197A or an approved equivalent. The mounting hood and camera shall be supplied by the contractor as part of the

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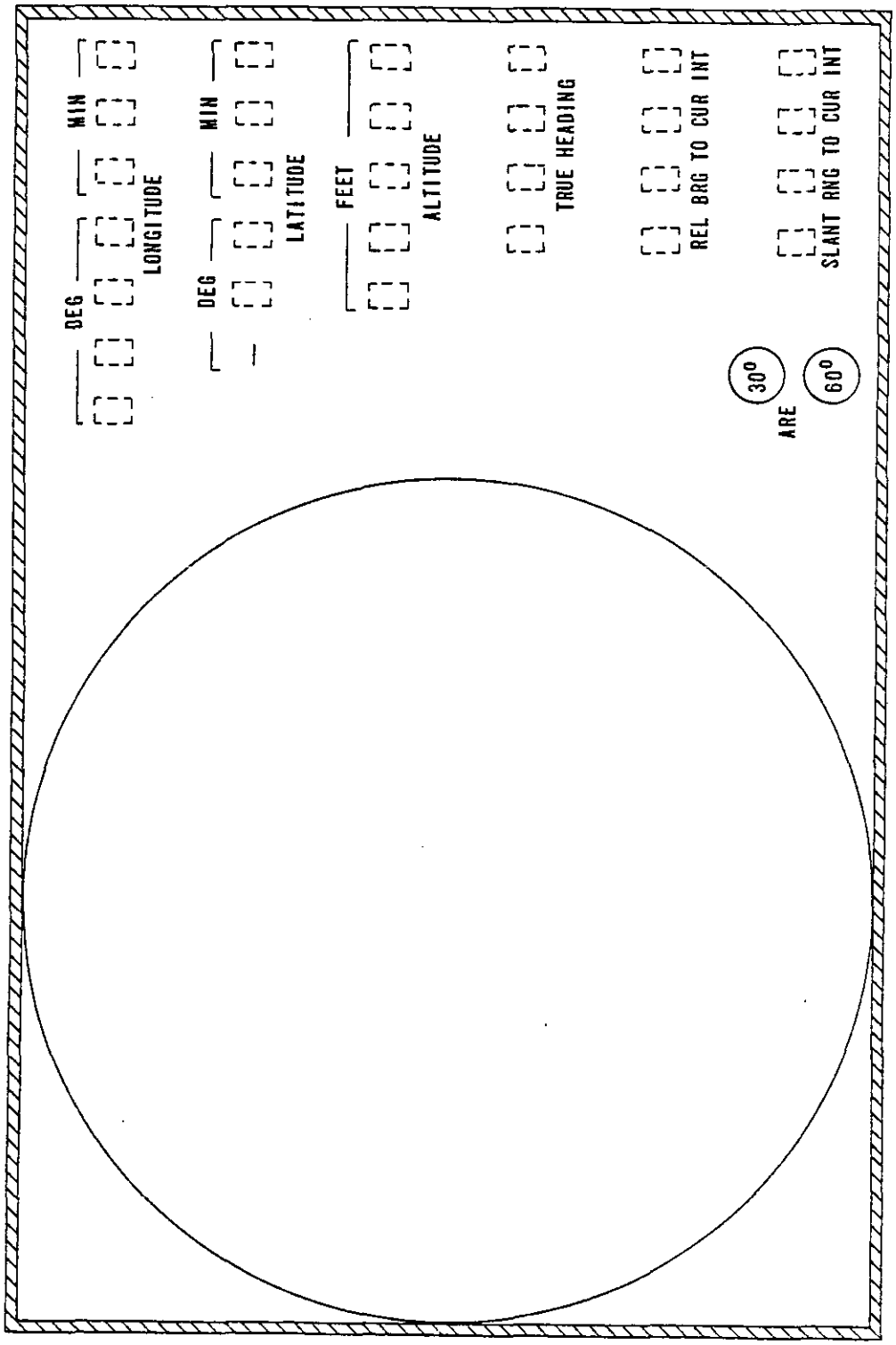


FIGURE 2. DISPLAY.

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TABLE I. Data for normal display.

	PRESENTATION	DISPLAY FORM AND UNITS
1	LONGITUDE	+/-XXX°XX.X' 1/
2	LATITUDE	+/-XX°XX.X' 2/
3	ALTITUDE	XXXXX FEET
4	TRUE HEADING	XXX.X° 3/
5	RELATIVE BEARING TO CURSOR INTERSECT	XXX.X° 3/
6	SLANT RANGE TO CURSOR INTERSECT	XXX.X NAUTICAL MILES
7	GROUND TRACK HEADING	XXX.X° 3/
8	GROUND TRACK VELOCITY	XXXX.X KNOTS
9	GROUND RANGE TO DESTINATION	XXX.X NAUTICAL MILES
10	TRUE HEADING TO DESTINATION	XXX.X° 3/
11	TIME TO DESTINATION	XXX. SECONDS
12	SPARE	XXXXXX
13	SPARE	XXXXXX
14	SPARE	XXXX
15	GROUND RANGE TO CURSOR INTERSECT	XXX.X NAUTICAL MILES

1/ +/- = E/W 2/ +/- = N/S

3/ Display Limits: 000.0° to 359.9°

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TABLE II. Data for freeze display.

SWITCH POSITION	DATA WORD	DISPLAY FORM
1	VERT. LOOP ERROR TERM	±XXXXX. Ft.
2	TRUE AIR SPEED	XXXX. fps
3	VERT. SEPER. TO DEST.	XXXXX(x10) Ft.
4	VEL. COR. /M. T. VEL.	XXXX. fps
5	WEAPON TIME TO FALL	XXX.X sec
6	TOTAL VELOCITY	XXXX. fps
7	XW RANGE TO TARGET	±XXXXX(x10) Ft.
8	COMP. BALLISTICS RANGE	±XXXXX(x10) Ft.
9	STEERING ERROR	±XX.X° 2/
10	TRUE HEADING	XXX.X° 1/
11	ATTACK MODE	±XXXXX(octal) 3/
12	NORMAL ACCELERATION	±X.XX g
13	RELEASE ADVANCE	±XXXXX(x10) Ft.
14	WIND VELOCITY	XXXX fps
15	WIND HEADING	XXX.X° 1/
16	ANGLE OF ATTACK	±XX.X° 2/
17	M.S.L. ALTITUDE	XXXXX Ft.
18	SPARE	
19	SPARE	
20	SPARE	

1/ Display Limits : 000.0° to 359.9°
2/ Display Limits : +89.9° to -90.0°
3/ Display Limits : -77777 to +77777; this display has no scale factor since it is a display of a binary word in octal representation

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equipment. The Camera and Hood is a detachable assembly and shall not be subject to the design and construction requirements of MIL-E-16400 as listed in paragraph 3.3.

*3.4.7 Built-In Test Equipment - The equipment shall be provided with an oscilloscope, type AN/USM-296 (or equivalent) and built-in-test circuits to enable the location of faults to a replaceable subassembly.

*3.4.8 Power Converter - The power converter shall provide 115 Volts, single phase, 400 Hz power to the Airborne Video Tape Recorder/Reproducer type AN/USH-17(V). The power converter shall be capable of providing 400 VA continuous and peak currents of 6 amperes RMS for 3 seconds. The power converter shall not be subject to the design and construction requirements of MIL-E-16400 as listed in paragraph 3.3.

3.5 Detailed Requirements -

3.5.1 Hard Copy Radar Display (A1) - The Hard Copy Radar Display shall meet the following requirements.

3.5.1.1 Function - The Hard Copy Radar Display shall provide the identical radar display and data block quantities displayed on the Radar Display (A4) and shall also contain the ICS audio speaker and amplifier.

3.5.1.2 Form Factor - The Hard Copy Radar Display shall be rack mounted and shall not exceed 8-3/4 inches high, 19 inches wide and 20 inches deep, excluding handles and control knobs.

3.5.1.3 Weight - The weight of the Hard Copy Radar Display shall not exceed 56 pounds.

*3.5.1.4 Contents - The Hard Copy Radar Display shall contain the below listed subassemblies and printed wiring assemblies (PWA's). PWA's shall be 5 inches by 6-1/4 inches with electrical interface terminated into a M21097/14-07 connector.

(1) Electron Tube Assy, Cathode Ray (A1A1) - The CRT shall be used as the radar display indicator. The yoke shall provide magnetic deflection of the electron beam.

(2) Indicator Assy, Digital Display (A1A2) - This subassembly shall display the data block quantities 1 through 6 of Table I.

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(3) Divider, High Voltage (A1A3) - This subassembly shall provide a means for measuring the CRT anode voltage. A voltage division of 1000 shall be used.

(4) Power Supply Assy, HV (A1A4) - This power supply shall provide 8500 volts at current up to 500 u amps to the radar display CRT. This subassembly shall be nonrepairable.

(5) Amplifier-Driver, Deflection Coil (A1A5) - This subassembly shall provide vertical deflection current to the deflection yoke of the radar display CRT. It shall be a linear voltage-to-current converter.

(6) Amplifier-Driver, Deflection Coil (A1A6) - This subassembly shall be identical to A1A5 except it shall provide the horizontal deflection current.

(7) Circuit Card Assembly, Video Amplifier (A1A7) - This PWA shall provide amplification and dc restoration of the input video signal.

(8) Circuit Card Assembly, Local-Remote Control (A1A8) - This PWA shall provide the capability of controlling the CONTRAST, BRIGHTNESS, VERTICAL, and HORIZONTAL position of the A1A1 CRT from either the local controls in A1 or remotely via the controls in the Radar Indicator, A4.

(9) Circuit Card Assembly, Unblanking and Phosphor Protect (A1A9) - This PWA shall provide unblanking of the radar display indicator, protect the indicator in the event of sweep loss or the loss of cathode bias, and it shall provide proper gating for the Automatic Range Expansion (ARE) mode indicator lights.

(10) Circuit Card Assembly, Audio Amplifier (A1A10)- This PWA is an audio power amplifier for the ICS audio.

(11) Circuit Card Assembly, Shutter Sync Horizontal Sweep Sampler (A1A11) - This PWA shall provide synchronization of the camera shutter with the beginning and ending of any radar left-to-right or right-to-left sweep.

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(12) Circuit Card Assembly, Shutter Sync Edge Detector and Logic (A1A12) - This PWA shall utilize the output signals from A1A11, and a shutter enabling signal to produce camera shutter control signals.

(13) Circuit Card Assembly, Fault Sensing (A1A13) - This PWA shall monitor selected circuit outputs during the TEST mode and shall energize a fault indicator light in the event that one of the selected outputs is absent.

(14) Circuit Card Assembly, Extender (A1A14) - This PWA shall provide electrical connection between an extended assembly and its chassis connector.

(15) Bias Supply (A1A15) - This subassembly shall provide focus voltage and screen voltage for the CRT and the voltage for A1A9.

(16) Chassis Assembly, Hard Copy Radar Display (A1A16) - This chassis assembly shall provide the mounting facilities and interconnecting wiring to interface the above subassemblies together and with the front panel controls and external cables.

(17) Camera, Still Picture, KD-44A (A1A17) - The Camera and Hood shall meet the following requirements:

a. Function - The camera shall provide hard copy photographs of the radar display and selected data quantities presented on A1. The hood shall provide the mechanical interface and prevent the impingement of ambient illumination on the face of the display panel.

b. Form Factor - The camera and hood shall mount on the front of A1. The camera shall be hinged on the hood to allow visual access of the display panel. The camera and hood shall not exceed 8-1/2 inches high, 11-1/2 inches wide and 17 inches deep.

c. Weight - The weight of the camera and hood shall not exceed 15 pounds.

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d. Storage Temperature - The camera shall provide hard copy photographs after storage (without film) within the temperature limits from -40°C to +75°C.

e. Operating Temperature - The camera (with film) shall provide hard copy photographs over the temperature range from -5°C to + 50°C.

f. Humidity - The camera (with film) shall provide hard copy photographs when operated in a relative humidity not to exceed 90% over the temperature range from -5°C to + 50°C.

g. Vibration - The camera shall not be damaged when subjected to vibration of 0.01 inch double amplitude over the frequency range from 10 to 55 Hz.

(18) Cable Assembly (W1) - This cable assembly provides electrical interconnection between the Camera, Still Picture, KD-44A (A1A17) and the front panel of Indicator, Radar IP-1100/UPQ-5 (A1).

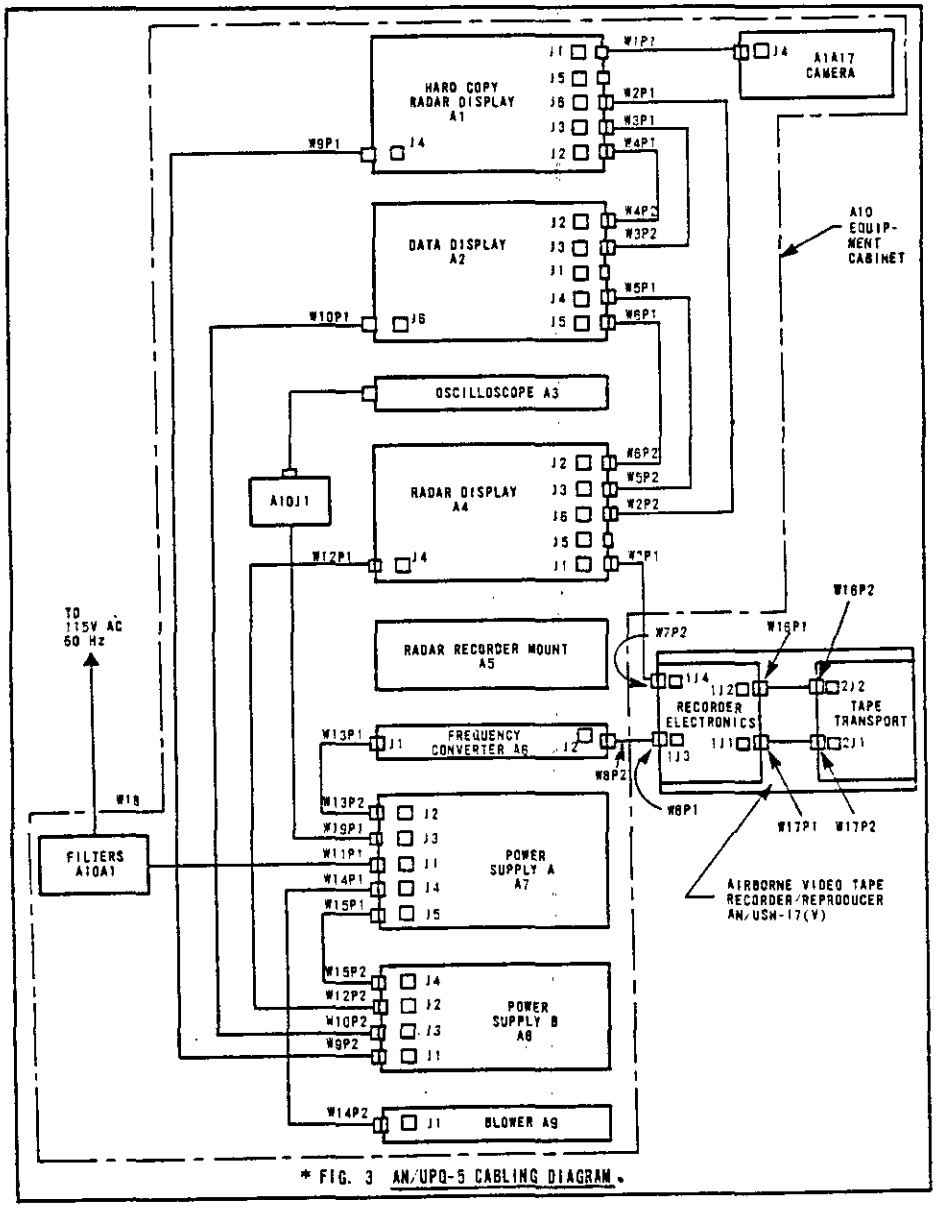
*3.5.1.5 Controls - The Hard Copy Radar Display shall have the front panel controls:

Control	Function
CONTRAST	Adjust display contrast
BRIGHTNESS	Adjust display brightness
VERTICAL	Position display vertically
HORIZONTAL	Position display horizontally
LOCAL-REMOTE	Selects A1 or A4 as source for the four controls above
AUDIO LEVEL	Adjust ICS audio level

*3.5.1.6 Electrical Connectors - Connections to external circuits shall be provided as follows (fig. 3):

Reference Designation	Receptacle Type	Function
J1	MS3120E-12-10S	Provide power and control for camera

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* FIG. 3 AN/UPQ-5 CABLING DIAGRAM.

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Reference Designation	Receptacle Type	Function
J2	MS3120E-20-41P	Radar display data
J3	MS3120E-20-41PW	Radar display data
J4	MS3120E-22-41P	Power
J6	L22TE95PONA 1/ (or an approved equivalent)	Signal from Radar Display

3.5.1.7 Test Points - The Hard Copy Radar Display shall have the test points shown on Table III.

3.5.2 Data Display (A2) - The Data Display shall meet the following requirements.

*3.5.2.1 Function - The Data Display decodes, processes and displays the aircraft digital data block and digital bombing solution quantities. The Data Display shall also provide the drive for the six quantities displayed on A1 and A4.

3.5.2.2 Form Factor - The Data Display shall be rack mounted and shall not exceed 7 inches high, 19 inches wide and 20 inches deep, excluding handles and control knobs.

3.5.2.3 Weight - The weight of the Data Display shall not exceed 40 pounds.

*3.5.2.4 Contents - The Data Display shall contain the below listed subassemblies. PWA's shall be 6-3/4 inches by 4-3/4 inches with electrical interface terminated into M21097/14-13 connectors.

(1) Control Panel Assembly (A2A1) - This assembly contains the Indicator Assembly, Digital Display (A2A1A1), the front panel controls, and the Data Freeze quantities identification plate.

(2) Circuit Card Assembly, Two-Digit Data Freeze Memory (A2A2, A2A3, A2A4) - Two digits of processed bombing target quantities (Data Freeze) are strobed into appropriate memory addresses where they are stored and read out as desired.

1/ Burndy Corp., Part Number.

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TABLE III. HARD COPY RADAR DISPLAY (A1) TEST POINTS.

REFERENCE DESIGNATION	DESCRIPTION
A1A3TP1	CRT ANODE VOLTAGE x 1/1000
A1A5TP1	VERT SWP INPUT
A1A5TP2	VERT POSITIONING INPUT
A1A5TP3	VERT SWP VOLTAGE TO DRIVER TRANSISTORS
A1A5TP4	VOLTAGE PROPORTIONAL TO VERT SWP CUR
A1A6TP1	HORIZ SWP INPUT
A1A6TP2	HORIZ POSITIONING INPUT
A1A6TP3	HORIZ SWP VOLTAGE TO DRIVER TRANSISTORS
A1A6TP4	VOLTAGE PROPORTIONAL TO HORIZ SWP CUR
A1A7TP1	VIDEO FAULT SENSE
A1A7TP2	BRIGHTNESS LEVEL
A1A9TP1	PHOSPHOR PROTECT GATING LEVEL
A1A9TP2	HIGH VOLTAGE POWER SUPPLY CONTROL
A1A9TP3	CRT BLANKING WAVEFORM
A1A10TP1	AUDIO VOLTAGE TO DRIVER TRANSISTOR
A1A12TP1	SHUTTER OPEN CONTROL
A1A12TP2	CAMERA SYNC
A1A12TP3	SHUTTER UNBLANKING
A1A12TP4	STORAGE PULSE
A1A13TP1	HIGH VOLTAGE SENSE OUTPUT
A1A13TP2	HORIZ SWP SENSE OUTPUT
A1A13TP3	BLANKING SENSE OUTPUT
A1A13TP4	CAMERA SYNC SENSE OUTPUT
A1A13TP5	VERT SWP SENSE OUTPUT
A1A13TP6	VIDEO SENSE OUTPUT
A1A15TP1	+110 V DC
A1A15TP2	FOCUS GRID VOLTAGE
A1A15TP3	SCREEN VOLTAGE
A1A15TP4	+450 V DC

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(3) Circuit Card Assembly, BCD Counter (A2A5) - This PWA accepts scaled binary data, in the form of a pulse train and converts into BCD for display.

(4) Circuit Card Assembly, Scaled Clock Control (A2A6) - This PWA selects the proper accuracy control gates for the scaled binary data pulse train.

(5) Circuit Card Assembly, Two-Digit Data Block Memory (A2A7, A2A8, A2A9) - Two digits of each navigational data quantity are strobed in memory. The memory is commutated at a 2 kHz rate causing a continuous readout of the navigational data quantities.

(6) Circuit Card Assembly, Clock Generator (A2A10) - This PWA is a 3 MHz crystal-controlled oscillator counted down to provide the required frequencies from 3 MHz to 125 Hz.

(7) Circuit Card Assembly, Data Freeze Control "C" (A2A11) - This PWA contains the logic functions pertinent to the control of the system.

(8) Circuit Card Assembly, Data Freeze Control "A" (A2A12) - This PWA contains the logic functions used to sequence through the address search, pre-count, processing and write modes of operation.

(9) Circuit Card Assembly, Sync Detector and Decoder (A2A13) - This PWA detects and decodes the ones and zeroes of the input signal; a shift clock is generated and the data is examined for validity and parity.

(10) Circuit Card Assembly, Shutter Control and Sign Storage (A2A14) - This PWA controls the length of time the navigational data is displayed for the hard copy printout. This PWA shall also store the signs of the latitude and longitude.

(11) Circuit Card Assembly, Readout Drivers (A2A15, A2A16, A2A17) - These PWA's shall convert the low level navigational data control signals into 9 volt drive signals for the LED displays.

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(12) Circuit Card Assembly, Data Freeze Control "B" (A2A18) - This PWA generates the data freeze control signals which select addresses and display positions by utilizing a clocked counter and 1-of-16 and 1-of-10 decoders.

(13) Circuit Card Assembly, Freeze Address Decoder and Comparator (A2A19) - This PWA shall consist of a matrix of diodes which shall generate fixed addresses which are compared to the input address for identification of the freeze data.

(14) Circuit Card Assembly, Stored Scale Factors "A" (A2A20) - This PWA gates and stores the scale factors "A" which are selected by A2A19 and A2A25.

(15) Circuit Card Assembly, Data and Address Storage (A2A21) - Decoded data which is shifted onto this PWA is strobed into storage for processing.

(16) Circuit Card Assembly, Selected Scale Factors (A2A22) - The stored scale factors control gates which select the proper number of binary counts generated by the binary counter.

(17) Circuit Card Assembly, Binary Counter and Fixed Scale Factors (A2A23) - The binary counter is turned on for a fixed number of counts determined by the scale factor gates. The binary counts are used to process the data and ultimately generate the scaled binary pulse train.

(18) Circuit Card Assembly, Data Storage and Transition Gates (A2A24) - The decoded data is shifted onto this PWA and strobed into storage. The stored data gates the leading edge transitions of the binary counter into a scaled binary pulse train.

(19) Circuit Card Assembly, Block Address Decoder and Comparator (A2A25) - This PWA shall consist of a matrix of diodes which shall generate fixed addresses which are compared to the input address for identification of the block data.

(20) Circuit Card Assembly, Data Block Control (A2A26) - This PWA contains a counter and 1-of-16 decoder which continuously runs, controlling the navigational data display.

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(21) Circuit Card Assembly, Stored Scale Factors "B" (A2A27) - This PWA gates and stores the scale factors "B" which are selected by A2A19 and A2A25.

(22) Circuit Card Assembly, Error Sensing (A2A28) - Four distinct signals are sensed for their presence by this PWA. The absence of any one of the signals shall cause the fault light to light.

(23) Circuit Card Assembly, Self Test (A2A29) - A self test waveform is generated in the proper format on this PWA. The address is continuously updated to exercise the data block and data freeze quantities.

(24) Circuit Card Assembly, Extender (A2A30) - This PWA shall provide electrical connection between an extended assembly and its chassis connector.

*3.5.2.5 Controls - The Data Display shall have the following front panel controls:

Control	Function
DISPLAY SELECTOR	Select Data Freeze quantity of Table II to be displayed.
RESET	"Unfreeze" Storage Register enabling it to accept new freeze data and turn off weapon release indicator lamp.

*3.5.2.6 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designation	Receptacle Type	Function
J6	MS3120E-20-16P	Power
J4	MS3120E-20-41SX	Radar Display data
J5	MS3120E-20-41SY	Radar Display data

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Reference Designation	Receptacle Type	Function
J2	MS3120E-20-41S	Hard Copy Radar Display data
J3	MS3120E-20-41SW	Hard Copy Radar Display data

3.5.2.7 Test Points - The Data Display shall have the test points shown on Table IV.

*3.5.3 Oscilloscope AN/USM-296 (A3) - The Oscilloscope shall meet the following requirements:

*3.5.3.1 Function - The oscilloscope shall be used in conjunction with the built-in-test equipment to maintain the equipment. The oscilloscope shall be a type AN/USM-296 or equivalent.

3.5.3.2 Form Factor - The oscilloscope shall be rack mounted and shall not exceed 5-1/4 inches high, 19 inches wide and 20 inches deep, excluding handles and knobs.

3.5.3.3 Weight - The oscilloscope shall not exceed 40 pounds.

3.5.4 Radar Display (A4) - The Radar Display shall meet the following requirements:

3.5.4.1 Function - The Radar Display shall serve as the operator's position. The OPERATE, LAMP TEST and TEST status switch shall be on this WRA. The Radar Display shall accept horizontal and vertical sweep signals, intensity gate, video and ICS audio from the airborne recorder/reproducer. This WRA shall amplify the sweep, video and unblanking signals and shall provide a radar display.

3.5.4.2 Form Factor - The Radar Display shall be rack mounted and shall not exceed 10-1/2 inches high, 19 inches wide and 20 inches deep, excluding handles and control knobs.

3.5.4.3 Weight - The weight of the Radar Display shall not exceed 56 pounds.

*3.5.4.4 Contents - The Radar Display shall contain the below listed subassemblies and PWA's. PWA's shall be 5 inches by 6-1/4 inches with electrical interface terminated into M21097/14-07 connectors.

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TABLE IV. DATA DISPLAY TEST POINTS (A2).

REFERENCE DESIGNATION	DESCRIPTION
A2A2TP1	DATA FREEZE 2^2 UNITS
A2A2TP2	DATA FREEZE 2^3 UNITS
A2A2TP3	DATA FREEZE 2^1 UNITS
A2A2TP4	DATA FREEZE 2^0 UNITS
A2A2TP5	DATA FREEZE 2^2 SIGN
A2A2TP6	DATA FREEZE 2^3 SIGN
A2A2TP7	DATA FREEZE 2^1 SIGN
A2A2TP8	DATA FREEZE 2^0 SIGN
A2A3TP1	DATA FREEZE 2^2 TENS
A2A3TP2	DATA FREEZE 2^3 TENS
A2A3TP3	DATA FREEZE 2^1 TENS
A2A3TP4	DATA FREEZE 2^0 TENS
A2A3TP5	DATA FREEZE 2^2 TEN THOUSAND
A2A3TP6	DATA FREEZE 2^3 TEN THOUSAND
A2A3TP7	DATA FREEZE 2^1 TEN THOUSAND
A2A3TP8	DATA FREEZE 2^0 TEN THOUSAND
A2A4TP1	DATA FREEZE 2^2 HUNDREDS
A2A4TP2	DATA FREEZE 2^3 HUNDREDS
A2A4TP3	DATA FREEZE 2^1 HUNDREDS
A2A4TP4	DATA FREEZE 2^0 HUNDREDS
A2A4TP5	DATA FREEZE 2^2 THOUSANDS
A2A4TP6	DATA FREEZE 2^3 THOUSANDS
A2A4TP7	DATA FREEZE 2^1 THOUSANDS
A2A4TP8	DATA FREEZE 2^0 THOUSANDS
A2A5TP1	2ND DIGIT 2^3
A2A5TP2	3RD DIGIT 2^3
A2A5TP3	4TH DIGIT 2^0
A2A5TP4	3RD DIGIT CLOCK ENABLE
A2A6TP1	(5TH DIGIT 2^3)'

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TABLE IV. DATA DISPLAY TEST POINTS (A2) (Cont'd).

REFERENCE DESIGNATION	DESCRIPTION
A2A6TP2	(6TH DIGIT 2^3)'
A2A6TP3	SCALED BINARY COUNT
A2A6TP4	3 COUNT CONTROL
A2A6TP5	÷4 CONTROL
A2A7TP1	DATA BLOCK 2^3 10,000
A2A7TP2	DATA BLOCK 2^1 100,000
A2A7TP3	DATA BLOCK 2^2 100,000
A2A7TP4	DATA BLOCK 2^0 10,000
A2A7TP5	DATA BLOCK 2^1 10,000
A2A7TP6	DATA BLOCK 2^0 100,000
A2A7TP7	DATA BLOCK 2^3 100,000
A2A7TP8	DATA BLOCK 2^2 10,000
A2A8TP1	DATA BLOCK 2^3 HUNDREDS
A2A8TP2	DATA BLOCK 2^1 THOUSANDS
A2A8TP3	DATA BLOCK 2^2 THOUSANDS
A2A8TP4	DATA BLOCK 2^0 HUNDREDS
A2A8TP5	DATA BLOCK 2^1 HUNDREDS
A2A8TP6	DATA BLOCK 2^0 THOUSANDS
A2A8TP7	DATA BLOCK 2^3 THOUSANDS
A2A8TP8	DATA BLOCK 2^2 HUNDREDS
A2A9TP1	DATA BLOCK 2^3 UNITS
A2A9TP2	DATA BLOCK 2^1 TENS
A2A9TP3	DATA BLOCK 2^2 TENS
A2A9TP4	DATA BLOCK 2^0 UNITS
A2A9TP5	DATA BLOCK 2^1 UNITS
A2A9TP6	DATA BLOCK 2^0 TENS
A2A9TP7	DATA BLOCK 2^3 TENS
A2A9TP8	DATA BLOCK 2^2 UNITS

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*TABLE IV. DATA DISPLAY TEST POINTS (A2) (Cont'd).

REFERENCE DESIGNATION	DESCRIPTION
A2A10TP1	(3 MHZ)'
A2A10TP2	1 KHZ CLOCK
A2A10TP3	10 KHZ CLOCK
A2A10TP4	25 KHZ CLOCK
A2A10TP5	12.5 KHZ CLOCK
A2A10TP6	100 KHZ CLOCK
A2A11TP1	(END OF COUNT RESET)'
A2A11TP2	GATED 100 KHZ
A2A11TP3	(±180° SHIFT)'
A2A11TP4	FREEZE SELECT
A2A11TP5	(WEAPONS RELEASE LOCKOUT)'
A2A11TP6	GATED 12.5 KHZ
A2A11TP7	(START FREEZE SEARCH)'
A2A12TP1	DISABLE BLOCK DECODER
A2A12TP2	DISABLE FREEZE DECODER
A2A12TP3	(BLOCK WRITE ENABLE)'
A2A12TP4	(FREEZE WRITE ENABLE)'
A2A12TP5	STORE DECODED SCALE FACTOR
A2A12TP6	ADDRESS SEARCH OR END OF COUNT
A2A13TP1	(SYNC)'
A2A13TP2	(COUNT 3)'
A2A13TP3	(RESET STORAGE)'
A2A13TP4	(SHIFT CLOCK)'
A2A13TP5	SIGN AND SERIAL DATA A
A2A13TP6	(CLEAR SHIFT REGISTER)'
A2A13TP7	DATA INPUT
A2A13TP8	(START ADDRESS SEARCH)'
A2A14TP1	STORED SIGN
A2A14TP2	BRIGHTNESS CONTROL

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*TABLE IV. DATA DISPLAY TEST POINTS (A2) (Cont'd)-

REFERENCE DESIGNATION	DESCRIPTION
A2A14TP3	LONG (-) SIGN
A2A14TP4	LAT (-) SIGN
A2A15	NONE
A2A16	NONE
A2A17	NONE
A2A18TP1	(STOP FREEZE SEARCH)'
A2A18TP2	FREEZE RAM ADDRESS E
A2A18TP3	CHIP SELECT A
A2A18TP4	CHIP SELECT B
A2A18TP5	(DATA FREEZE CONTROL 15)'
A2A18TP6	FREEZE RAM ADDRESS D
A2A19TP1	FREEZE COMPARE A
A2A19TP2	FREEZE COMPARE B
A2A19TP3	FREEZE COMPARE C
A2A19TP4	FREEZE COMPARE D
A2A20TP1	STORED SCALE FACTOR 5,120
A2A20TP2	STORED SCALE FACTOR 65,536
A2A20TP3	STORED SCALE FACTOR 256
A2A20TP4	STORED SCALE FACTOR 108,000
A2A20TP5	STORED SCALE FACTOR 104,856
A2A20TP6	STORED SCALE FACTOR 4,096
A2A20TP7	STORED SCALE FACTOR FLAG
A2A21TP1	(LSB SCALED DATA)'
A2A21TP2	STORED ADDRESS 2^2
A2A21TP3	STORED ADDRESS 2^1
A2A21TP4	STORED ADDRESS 2^0
A2A21TP5	STORED ADDRESS 2^3
A2A21TP6	13TH SHIFT BIT

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*TABLE IV. DATA DISPLAY TEST POINTS (A2) (Cont'd).

REFERENCE DESIGNATION	DESCRIPTION
A2A22TP1	SELECTED SCALE FACTORS
A2A22TP2	256 GATE
A2A22TP3	SPARE GATE
A2A22TP4	5,120 GATE
A2A22TP5	104,856 GATE
A2A22TP6	6,900 GATE
A2A23	NONE
A2A24TP1	STORED 2^{-10}
A2A24TP2	SCALED DATA
A2A25TP1	BLOCK COMPARE A
A2A25TP2	BLOCK COMPARE B
A2A25TP3	BLOCK COMPARE C
A2A25TP4	BLOCK COMPARE D
A2A26TP1	BLOCK RAM ADDRESS A3
A2A27TP1	STORED SCALE FACTOR SPARE
A2A27TP2	STORED SCALE FACTOR 97,008
A2A27TP3	STORED SCALE FACTOR 3,200
A2A27TP4	STORED SCALE FACTOR 7,200
A2A27TP5	STORED SCALE FACTOR 14,400
A2A27TP6	STORED SCALE FACTOR 6,900
A2A27TP7	STORED SCALE FACTOR SPARE
A2A28	NONE
A2A29TP1	SELF TEST 1 KHZ
A2A29TP2	SELF TEST RZ SERIAL OUTPUT

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(1) Electron Tube Assy, Cathode Ray (A4A1) - Identical to A1A1.

(2) Indicator Assy, Digital Display (A4A2) - Identical to A1A2.

(3) Divider, High Voltage (A4A3) - Identical to A1A3.

(4) Power Supply Assy, HV (A4A4) - Identical to A1A4.

(5) Amplifier-Driver, Deflection Coil (A4A5) - Identical to A1A5.

(6) Amplifier-Driver, Deflection Coil (A4A6) - Identical to A1A6.

(7) Circuit Card Assembly, Video Amplifier (A4A7) - Identical to A1A7.

(8) Circuit Card Assembly, Unblanking and Phosphor Protect (A4A9) - Identical to A1A9.

(9) Circuit Card Assembly, Test Signal Generator (A4A12) - This PWA shall generate audio, unblanking, video, horizontal sweep and vertical sweep wave forms in the TEST mode of operation.

(10) Circuit Card Assembly, Fault Sensing (A4A13) - Identical to A1A13.

(11) Circuit Card Assembly, Extender Card A4A14) - Identical to A1A14.

(12) Bias Supply (A4A15) - Identical to A1A15.

(13) Chassis Assy, Radar Display (A4A16) - This chassis assembly shall provide the mounting facilities and interconnecting wiring to interface the above subassemblies together and with the front panel controls.

*3.5.4.5 Controls - The Radar Display shall have the following front panel controls:

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Controls	Function
CONTRAST	Adjust display contrast
BRIGHTNESS	Adjust display brightness
VERTICAL	Position display vertically
HORIZONTAL	Position display horizontally
STATUS	Select mode of operation; OPERATE, LAMP TEST and TEST
CAMERA SHUTTER	Operates camera shutter for hard copy

*3.5.4.6 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designation	Receptacle Type	Function
J1	48-10G20-28PN-(135)[2/] (or an approved equivalent)	Recorder inputs
J2	MS3120E-20-41P	Radar display data
J3	MS3120E-20-41PX	Radar display data
J4	MS3120E-22-41P	Power
J6	L22TE95SONA [3/] (or an approved equivalent)	Signals to Radar Display
J7	JJ-089	Audio output
J8	MS35179-1094A(BNC)	Radar video
J9	MS35179-1094A(BNC)	TV video

3.5.4.7 Test Points - The Radar Display shall have the test points shown on Table V.

3.5.5 Radar Recorder Mount (A5) - The Radar Recorder Mount shall meet the following requirements:

*3.5.5.1 Function - The Radar Recorder Mount shall support, position, and provide electrical connection for the Electronics Unit and the Transport Unit of the AN/USH-17(V) Airborne Video Tape Recorder/ Reproducer for playback into the equipment.

2/ Amphenol-Connector Div. The Bunker-Ramo Corp. part number

3/ Burndy Corp. part number

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TABLE V. RADAR DISPLAY (A4) TEST POINTS.

REFERENCE DESIGNATION	DESCRIPTION
A4A3TP1	CRT ANODE VOLTAGE x 1/1000
A4A5TP1	VERT POSITIONING INPUT
A4A5TP2	VERT SWP INPUT
A4A5TP3	VERT SWP VOLTAGE TO DRIVER TRANSISTOR
A4A5TP4	VOLTAGE PROPORTIONAL TO VERT SWP CUR
A4A6TP1	HORIZ POSITIONING INPUT
A4A6TP2	HORIZ SWP INPUT
A4A6TP3	HORIZ SWP VOLTAGE TO DRIVER TRANSISTORS
A4A6TP4	VOLTAGE PROPORTIONAL TO HORIZ SWP CUR
A4A7TP1	VIDEO FAULT SENSE
A4A7TP2	BRIGHTNESS LEVEL
A4A9TP1	PHOSPHOR PROTECT GATING LEVEL
A4A9TP2	HIGH VOLTAGE POWER SUPPLY CONTROL
A4A12TP1	AUDIO AND INTENSITY GATE TEST WAVEFORM
A4A12TP2	15.6 KHZ SQUARE WAVE
A4A12TP3	HORIZ SWP TEST WAVEFORM
A4A12TP4	VIDEO TEST WAVEFORM
A4A12TP5	VERT SWP TEST WAVEFORM
A4A13TP1	HIGH VOLTAGE SENSE OUTPUT
A4A13TP2	HORIZ SWP SENSE OUTPUT
A4A13TP3	BLANKING SENSE OUTPUT
A4A13TP4	CAMERA SYNC SENSE OUTPUT
A4A13TP5	VERT SWP SENSE OUTPUT
A4A13TP6	VIDEO SENSE OUTPUT
A4A15TP1	+110 V DC
A4A15TP2	FOCUS GRID VOLTAGE
A4A15TP3	SCREEN VOLTAGE
A4A15TP4	+450 V DC

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*3.5.5.1.1 Operational Access - The Transport Unit shall be positioned so that the tape can be changed without disconnecting the unit from the mount.

3.5.5.2 Form Factor - The Radar Recorder Mount shall be rack mounted and shall not exceed 10-1/2 inches high, 19 inches wide and 20 inches deep, excluding fastening hardware.

3.5.5.3 Weight - The weight of the Radar Recorder Mount shall not exceed 34 pounds.

*3.5.5.4 Contents - The Radar Recorder Mount shall include the subassemblies listed below:

(1) Chassis Assy, Recorder Mount (A5A1) - This chassis assembly shall provide the mechanical interface for mounting the AN/USH-17(V) Airborne Video Tape Recorder/Reproducer.

(2) Cable Assy, (W16) - This cable assembly interconnects the electronics package and the tape transport modules of the AN/USH-17(V).

(3) Cable Assy (W17) - This cable assembly interconnects the electronics package and the tape transport modules of the AN/USH-17(V).

*3.5.5.5 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designator	Receptacle Type	Function
W7P2	48-16G20-28PN-102(135)4/ (or an approved equivalent)	Signals out
W8P1	MS24266R12B35NX	Power
W17P1	48-16G22-32P-(172) 4/	Sec. 3.5.5.4

4/ Amphenol-Connector Div. The Bunker-Ramo Corp. part number

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Reference Designator	Receptacle Type	Function
W17P2	48-16G22-32S-(172)[5/]	Sec. 3.5.5.4
W16P1	48-16G24-43P-(172)[5/]	Sec. 3.5.5.4
W16P2	48-16G24-43S-(172)[5/]	Sec. 3.5.5.4

3.5.6 Power Converter (A6) - The Power Converter shall meet the following requirements:

3.5.6.1 Function - The Power Converter shall convert 60 Hz power to provide 400 va at 113 ± 5 vac, 400 ± 20 Hz, to a steady state load having a power factor in the range from 0.92 lead to 0.65 lag.

3.5.6.2 Form Factor - The Power Converter shall be rack mounted and shall not exceed 3-1/2 inches high, 19 inches wide and 19 inches deep.

3.5.6.3 Weight - The weight of the Power Converter shall not exceed 65 pounds.

*3.5.6.4 Contents - The Power Converter shall contain the following subassemblies:

(1) Control Regulator Board Subassembly - Regulates and controls DC output voltage.

(2) Inverter Driver Circuit Board Subassembly - Generates single phase, 400 Hz, AC voltage.

(3) Filter Subassembly - Provides input RFI filtering and filtering of the power output. This subassembly shall be nonrepairable.

3.5.6.5 Controls - The power Converter shall have the following front panel switches:

[5/] Amphenol Connector Div. The Bunker-Ramo Corp. part number

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Switch	Function
OUTPUT POWER	Applies output power
INPUT POWER	Applies input power

*3.5.6.6 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designator	Receptacle Type	Function
J1	MS3102A-18-10P	Input power
J2	MS3102A-18-10S	Output power

3.5.6.7 Test Points - Front panel test points shall be provided to measure the output voltage.

3.5.6.8 Circuit Breakers - A circuit breaker shall be provided on the input of the Power Converter.

3.5.7 Power Supply A (A7) - Power Supply A shall meet the following requirements:

*3.5.7.1 Function - Power Supply A shall serve as a distribution point for 115 volt AC to the Blower, Power Converter, Oscilloscope and Power Supply B. It also contains the +32 VDC and -32 VDC power supplies and the main power circuit breaker/power switch.

3.5.7.2 Form Factor - Power Supply A shall be rack mounted and shall not exceed 8-3/4 inches high, 19 inches wide and 20 inches deep.

3.5.7.3 Weight - The weight of Power Supply A shall not exceed 119 pounds.

*3.5.7.4 Contents - Power Supply A shall contain the following subassemblies:

(1) Chassis Assy, Power Supply A (A7A1) - The Power Supply A chassis assembly shall provide mounting facilities and interconnecting wiring for the ±32V power supplies, front panel controls, and external cables.

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(2) Power Supply Subassy, 32 VDC (A7PS1, A7PS2) - A +32 volt DC supply capable of supplying 6.0 amperes, and a -32 volt DC supply capable of supplying 6.0 amperes.

*3.5.7.5 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designator	Receptacle Type	Function
J1	L22TE19PONA 6/ (or an approved equivalent)	Input power
J2	MS3120E-12-3S	Power to the Power Converter (A6)
J3	MS3120E-12-3S	Power to the Oscilloscope (A3)
J4	MS3120E-12-3S	Power to the Blower (A9)
J5	MS3120E-22-41S	Power to Power Supply B(A8)

*3.5.7.6 Test Points - Front panel test points shall be provided to measure the output voltage of the +32 and -32 volt DC supplies and the AC voltage being supplied to the blowers.

3.5.7.7 Circuit Breakers - A circuit breaker shall be provided on the input of each power supply.

3.5.8 Power Supply B (A8) - Power Supply B shall meet the following requirements:

3.5.8.1 Function - Power Supply B is the distribution point for all the power required by the Hard Copy Radar Display, Radar Display and Data Display, excluding high voltage requirements.

3.5.8.2 Form Factor - Power Supply B shall be rack mounted and shall not exceed 7 inches high, 19 inches wide and 20 inches deep.

6/Burndy Corp., part number.

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3.5.8.3 Weight - The weight of Power Supply B shall not exceed 99 pounds.

*3.5.8.4 Contents - Power Supply B shall contain the following nonrepairable subassemblies:

(1) Chassis Assy, Power Supply B (A1) - This chassis assembly provides mounting and interconnecting wiring between the following listed subassemblies, the front panel, and the external cables.

(2) Power Supply Subassy, +450 VDC (PS1) - A 450 Volt DC supply capable of supplying 0.1 ampere.

(3) Power Supply Subassy, ±15 VDC (PS2) - A ±15 Volt DC supply capable of supplying 0.5 ampere.

(4) Power Supply Subassy, +10 VDC (PS3) - A 10 Volt DC supply capable of supplying 6.0 amperes.

(5) Power Supply Subassy, +6 VDC (PS4) - A 6 Volt DC supply capable of supplying 12.0 amperes.

(6) Overvoltage Protector Subassy, +10V (Z1) - Provides overvoltage protection on the output of the 10 Volt DC supply.

(7) Overvoltage Protector Subassy, +6V (Z2) - Provides overvoltage protection on the output of the 6 Volt DC supply.

*3.5.8.5 Electrical Connectors - Connectors to external circuits shall be provided as follows (fig. 3):

Reference Designators	Receptacle Type	Function
J1	MS3120E-22-41SX	Voltages to Hard Copy Radar Display (A1)
J2	MS3120E-22-41S	Voltages to Radar Display (A4)

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Reference Designators	Receptacle Type	Function
J3	MS3120E-20-16S	Voltages to Data Display (A2)
J4	MS3120E-22-41P	Power from Power Supply A (A7)

3.5.8.6 Test Points - Front panel test points shall be provided to measure the output voltage of each power supply.

3.5.8.7 Circuit Breakers - A circuit breaker shall be provided on the input of each power supply.

3.5.9 Blower (A9) - The Blower shall meet the following requirements:

3.5.9.1 Function - The blower shall provide up to 290 CFM of circulating cooling air for the equipment.

3.5.9.2 Form Factor - The Blower shall be EMI shielded, rack mounted and shall not exceed 6-1/8 inches high, 19 inches wide or 13 inches deep.

3.5.9.3 Weight - The weight of the blower shall not exceed 25 pounds.

3.5.9.4 Blower Power Requirements - The blower shall operate on 115 volts, single phase, 50/60 Hz at a nominal running current of 1.3 amperes.

*3.5.10 Equipment Cabinet (A10) - The Equipment cabinet shall meet the following requirements:

*3.5.10.1 Function - The Equipment cabinet shall provide an enclosure, mounting facilities, and interconnecting cabling for the WRA's of the Display Set, Mission Recorder AN/UPQ-5B.

*3.5.10.2 Form Factor - The Equipment cabinet shall be a ruggedized, low carbon steel unit and shall not exceed 24 inches wide, 27 inches deep and 72 inches high, excluding lifting eyes and casters.

*3.5.10.3 Weight - The weight shall not exceed 400 pounds.

*3.5.10.4 Contents - The Cabinet, Electrical Equipment CY-7319A/UPQ-5 shall contain the following:

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(1) RFI Filter Assy - The RFI Filter Assembly and its associated cable assy (W18) provides RFI filtering on the input power line. The RFI Filter Assembly contains (2) double L filters, weighs no more than 12 pounds, and its size shall not exceed 3 inches high, 6 inches wide, and 12 inches deep.

(2) Cable Assemblies - The Equipment Cabinet, contains the following cable assemblies which interconnect the WRA's: W2, W3, W4, W5, W6, W7, W8, W9, W10, W11, W12, W13, W14, W15, and W19 (fig. 3).

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of specification MIL-E-16400.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the

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preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4. Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated in all modes long enough to permit the equipment temperature to stabilize and to check sufficient characteristics in accordance with Section 3 of this specification, and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 10 to 30 cps (excluding resonant points) and a minimum double amplitude of 0.02 inch.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1 7/
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

*4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34. Any failure occurring in the camera, oscilloscope or time totalizing meters during reliability tests shall be considered non-relevant failures when determining the MTBF of the equipment.

*4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in

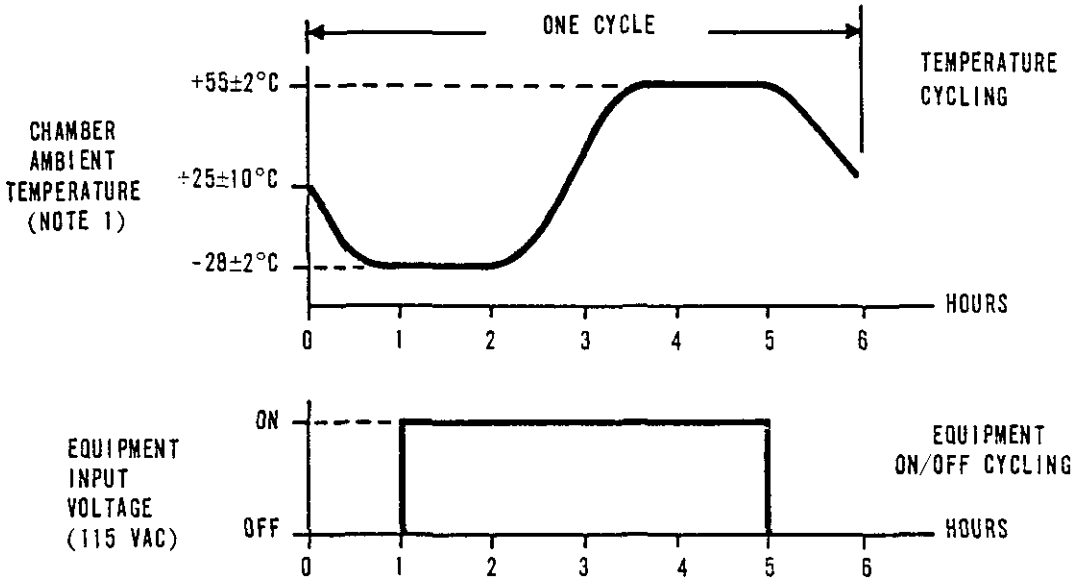
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7/ One out of first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

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Table 5 of MIL-STD-781. For the Qualification Phase, the Test level shall be as specified herein for the All-Equipment Test. The Accept-Reject criteria for Test Plan III shall be used.

*4.4.3.2 Reliability Production Acceptance Phase Tests - All equipments produced shall be subjected to the All-Equipment Test as specified herein. Test Level E of MIL-STD-781, as modified herein in accordance with Figure 4, shall be used.



NOTE 1: THE TEMPERATURE CONTROL IS PROGRAMMED TO CHANGE AMBIENT TEMPERATURE AT THE 2 AND 5 HOUR POINTS AND TO MAINTAIN CONSTANT TEMPERATURE BETWEEN THE 1 TO 2 AND 4 TO 5 HOUR POINTS OF THE TEST CYCLE.

FIG. 4. ALL-EQUIPMENT TEST CYCLE.

*4.4.3.2.1 All-Equipment Test - Each equipment produced shall be tested to accumulate an equipment "ON" time of 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the

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details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision. Any corrective action required as a result of a reject condition shall be handled in accordance with par. 5.7 of MIL-STD-781B.

*4.4.3.2.1.1 The all-equipment test cycle consists of vibration and temperature cycling, and cycling the equipment between its non-operating and operating states. Vibration cycling is performed with the equipment in the cabinet in its operating state and with vibration applied in a direction perpendicular to the equipment base. Temperature cycling is performed on WRA'S A1 through A9 (listed in paragraph 1.2), which may be removed from the equipment cabinet and installed in a test fixture which fits into the temperature chamber if the temperature chamber is of insufficient size to hold the entire cabinet. The Digital Display Indicator subassemblies in WRA'S A1, A2, and A4 will not be subjected to temperature cycling. They will be removed from the WRA'S and physically mounted outside the temperature chamber. Reconnection to their respective WRA'S will be accomplished via an extender cable permitting normal electrical operation. The WRA'S are interconnected by a test harness and the test fixture is designed to allow adequate chamber cooling air flow from the equipment blower (WRA 9) to the other WRA'S. During temperature cycling, the equipment is also cycled between its non-operating and operating states.

*4.4.3.2.1.2 Prior to starting the all-equipment test cycle a vibration survey is performed on one equipment to select a single non-resonant frequency, between 16 and 33 hertz, to be used during the all-equipment test vibration cycling. The survey is performed with vibration applied in a direction perpendicular to the equipment base. The vibration table displacement amplitude (1/2 total excursion) is 0.020 ± 0.004 for frequencies between 16 and 25 hertz and 0.010 ± 0.002 for frequencies between 25 and 33 hertz.

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*4.4.3.2.1.3 The all-equipment test cycle for each equipment is described by the following steps:

(1) Vibrate operating equipment for 60, +6 -0 minutes in direction and at vibration level and frequency selected in vibration survey.

(2) Subject WRA'S A1 through A9 to 6 consecutive cycles of temperature and equipment operation as shown in Figure 4.

(3) Install WRA'S into equipment cabinet and repeat step (1).

(4) Repeat step (2).

(5) Repeat step (3).

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

*4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity and the contractor.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the production tests shall be used where applicable. When not applicable, the contractor shall prepare a

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test procedure and submit it to the procuring activity for approval prior to conducting the tests.

*4.4.5 Equipment Failure - Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) (Test time accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

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Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The

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procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the Contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - The Mission Recorder Display Set shall faithfully display pulse radar ground map presentations, digital data block information and reproduce audio signals for Navy operational use in both shipboard and shore installations. As such, the design should insure good reliability and maintainability in a fleet environment. It is intended that the surface based display system shall be maintained by avionics personnel.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification. Data requirements will be identified in the contract rather than by specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from

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the use of materials, parts and processes other than those specified in Specification MIL-E-16400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

(1) Airborne Video Tape Recorder/Reproducer, AN/USH-17(V)

(2) Hood Assembly, FSN IR 1560 938 9807FA

*6.8 This specification is under the cognizance of AIR-533322A

Preparing Activity:
(Navy-AS)
Project No. N183

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-81688A(AS) Display Set, Mission Recorder AN/UPQ-5B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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